

Heart Failure Among 173,000 Community-Dwelling Participants From 25 Low-, Middle-, and High-Income Countries in the PURE Study

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ABSTRACT

BACKGROUND Most population studies examining heart failure (HF) have been conducted in Western high-income countries (HICs), with limited comparable data from lower-income settings.

OBJECTIVES The aims of this study were to describe differences in HF incidence and 30-day, 1-year, and 5-year case fatality rates among HF patients from countries at different income levels and in different global regions and to examine the impact of common and potentially modifiable risk factors for incident HF.

METHODS This analysis of the PURE (Prospective Urban Rural Epidemiology) study included 172,653 individuals from 25 HICs, upper middle-income countries (UMICs), lower middle-income countries (LMICs), and low-income countries (LICs) and 8 geographic regions of the world, followed for a median of 15 years. Age- and sex-standardized HF incidence, as well as 30-day, 1-year, and 5-year HF case fatality, were compared by income group and by geographic region. The population attributable fractions (PAFs) for incident HF related to 13 cardiometabolic, lifestyle, socioeconomic, environmental, and psychosocial risk factors were also estimated.

RESULTS The standardized rate of incident HF was 0.39 (95% CI: 0.36-0.41) per 1,000 person-years overall; the rate was highest in UMICs (0.58; 95% CI: 0.52-0.64), followed by HICs (0.36; 95% CI: 0.30-0.43), then LMICs (0.34; 95% CI: 0.30-0.38), and then LICs (0.26; 95% CI: 0.22-0.30). Among regions, the highest HF incidence was in sub-Saharan Africa (1.18; 95% CI: 0.95-1.41) and Europe and Central Asia (0.86; 95% CI: 0.72-1.00) and lowest in South Asia (0.19; 95% CI: 0.15-0.22). Thirty-day case fatality was highest in LICs (59%) and lowest in HICs (11%); it was highest in South Asia (63%) and sub-Saharan Africa (63%) and lowest in North America (12%). Five-year case fatality after HF diagnosis was highest in LICs (77%) and lowest in HICs (28%); it was highest in South Asia (81%) and sub-Saharan Africa (75%) and lowest in North America (25%). More than 71% of the PAF for HF was attributable to the 13 modifiable risk factors studied, the largest being hypertension (PAF = 25%).

CONCLUSIONS HF incidence and associated mortality vary substantially across countries at different levels of economic development and by geographic region. Hypertension is the largest population-level risk factor for HF globally. Preventive measures, early diagnosis, and access to guideline-directed medical therapy should be prioritized to reduce global disparities in HF incidence and mortality. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

ABBREVIATIONS
AND ACRONYMS

CVD = cardiovascular disease
HF = heart failure
HIC = high-income country
LIC = low-income country
LMIC = lower middle-income country
LVEF = left ventricular ejection fraction
MIC = middle-income country
PAF = population attributable fraction
UMIC = upper middle-income country

Hear failure (HF) is associated with high mortality, poor quality of life, and high health care costs.¹ Incidence increases steeply with age, and with rising life expectancy globally, the number of people with HF will grow. Data on the prevalence and incidence of HF are derived mostly from Western high-income countries (HICs), with comparatively fewer data for other populations.² There is a need for comparative data to better understand whether HF incidence and outcomes in middle-income countries (MICs) and low-income countries (LICs) and outside North America and Europe substantially differ from what is known in Western HICs. Studies detailing the predominant risk factors for HF development have also been focused largely on populations from HICs.^{2,3} Given that both the underlying causes of HF and the prevalence of its risk factors vary globally, there is a need to examine risk factors for HF in studies with greater representation from MICs and LICs.⁴

The PURE (Prospective Urban and Rural Epidemiology) study prospectively records information on the incidence of HF, risk factors, and outcomes in a

large contemporary multinational cohort of community-dwelling adults from 25 HICs, MICs, and LICs.^{5,6} In this analysis of the PURE study, our primary objectives were to describe differences in HF incidence and in 30-day, 1-year, and 5-year case fatality rates among HF patients from countries at different income levels and in different regions of the world. We also examined the comparative impact of 13 common and potentially modifiable clinical cardiovascular risk factors for incident HF in this global cohort.

METHODS

STUDY DESIGN AND PARTICIPANTS. The design of the PURE study has been reported previously.^{7,8} In brief, the PURE study is an ongoing prospective, population-based cohort study that was designed to include countries from a wide range of economic and social circumstances, with a proportionally larger representation from MICs and LICs. Study enrollment occurred in 3 phases between January 2000 and 2019 from 25 countries on 5 continents. In the first phase of the PURE study, enrollment occurred in 17 countries between 2000 and 2017; in the second phase, enrollment occurred in 4 additional countries

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between 2012 and 2017; and in the third phase, enrollment occurred in 4 additional countries between 2015 and 2019. Follow-up occurs every 3 years and is ongoing. We grouped countries and territories by income level (ie, HICs, upper middle-income countries [UMICs], lower middle-income countries [LMIC] and LICs) on the basis of gross national income per capita according to the World Bank classification at the time of study inclusion. There were 4 HICs (Canada, Saudi Arabia, Sweden, and the United Arab Emirates), 10 UMICs (Argentina, Brazil, Chile, Ecuador, Kazakhstan, Malaysia, Poland, Russia, South Africa, and Turkey), 6 LMICs (China, Colombia, Kyrgyzstan, Iran, the West Bank, and the Philippines), and 5 LICs (Bangladesh, India, Pakistan, Tanzania, and Zimbabwe) ([Supplemental Table 1](#)). We further grouped countries and territories by 8 geographic regions (North America, South America, the Middle East, Europe and Central Asia, sub-Saharan Africa, South Asia, Southeast Asia, and China) ([Supplemental Table 1](#)). We grouped countries according to the World Bank geographic regions with the exception of China and Southeast Asia, which were individually grouped because of sufficient numbers of participants for analyses in these subregions.⁹

The study enrolled individuals 35 to 70 years of age. In each country, we recruited individuals from urban and rural communities on the basis of pre-specified criteria ([Supplemental Appendix A1](#)). Within each community, we selected households and individuals using sampling strategies that would minimize biases in the selection of individuals that could potentially influence associations between risk factors and outcomes. We aimed for a broadly representative sample of the sociodemographic composition of communities.⁷ Prior analyses of the first phase of PURE demonstrate that demographic and mortality data collected in PURE largely correlated with national data from the participating countries.⁸ For the present analysis, we included individuals from the 3 study phases who had completed at least 1 cycle of follow-up. For the analysis of incident HF, we excluded participants who reported an HF diagnosis at study enrollment.

The PURE study was approved by local ethics committees at each center, and all participants provided written informed consent.

PROCEDURES. The PURE study has used standardized case report forms for the recording of participant information, history of HF, and individual-, household-, and community-level risk factors at baseline. HF prevalence estimates are based on self-report at

study enrollment. Following enrollment, we contacted participants at least every 3 years to ascertain their vital status and to record information on the occurrence of specific events and hospital admissions. Major cardiovascular events, including HF and death, were adjudicated centrally in each country by trained physicians who used standardized definitions and case report forms. HF events (fatal or nonfatal) required reported present typical signs (eg, ankle edema, rales, jugular venous pressure elevation) or symptoms (eg, paroxysmal nocturnal dyspnea, dyspnea at rest, ankle edema), as further detailed in [Supplemental Appendix A2](#). Adjudicators classified an event as definite, probable, or possible according to available supporting documentation ([Box 1](#)). The distribution of definite, probable, and possible HF in different country income groups is presented in [Supplemental Figure 1](#). For the present analysis, an event was considered fatal if death occurred within 30 days of first HF presentation. In addition, using all available information, we used an algorithm to attribute an etiology to each HF event ([Supplemental Appendix A3](#)). We also performed a detailed review of supporting documentation to collect left ventricular function when clinically available. For this report, 12 risk factors were selected on the basis of their associations with incident cardiovascular disease (CVD) in prior studies,¹⁰ grouped into cardiometabolic risk factors (hypertension, diabetes, high non-high-density lipoprotein cholesterol, and abdominal obesity according to sex-specific waist-to-hip ratio thresholds of >0.9 in men and 0.85 in women), behavioral risk factors (tobacco use, alcohol use, diet quality, and physical activity), low education, household air pollution, low physical strength (grip strength), and depression. A 13th risk factor, self-reported ischemic heart disease, was also considered given its established association with incident HF (see [Supplemental Appendix A4](#) for definitions). Data checks, follow-up for data completeness at each round of follow-up, and analyses were performed at the Population Health Research Institute.

OUTCOMES. The primary outcome was a first fatal or nonfatal HF event. We considered all reported first unrefuted HF events. Secondary outcomes were death within 30 days, 1 year, and 5 years after a first HF event.

STATISTICAL ANALYSIS. Continuous variables are expressed as mean $\pm$ SD and categorical variables as frequency counts and proportions. Between-group differences were tested using independent Student's *t*-tests or the Mann-Whitney *U* test for

BOX 1 HF Definition and Classification in PURE

HF classification	
Fatal first HF event	Defined as death within 30 d of first HF presentation in this analysis
Adjudication classifications ^a	
Definite	Signs ^b or symptoms ^c and radiologic signs of pulmonary congestion and/or ongoing diuretic treatment
Probable	Progressive dyspnea lying down or at night improving on sitting up and signs or symptoms of congestion ^d with known history of hypertension or MI/angina/other heart disease
Possible	Progressive dyspnea lying down or at night improving sitting up and signs or symptoms of congestion ^d

^aClassification of adjudicated HF events is further detailed in [Supplemental Appendix A2](#). ^bRales, increased jugular venous pressure, or ankle edema. ^cNocturnal paroxysmal dyspnea, dyspnea at rest, or ankle edema. ^dSwelling of feet, distention of abdomen, progressive cough.

HF = heart failure; PURE = Prospective Urban Rural Epidemiology.

continuous variables and the chi-square test for categorical variables. We calculated standardized incidence rates of first HF events using the age and sex distribution of the PURE population. Events recorded until July 3, 2025, were included in the present analysis. Incidence rates are expressed as events per 1,000 person-years. Time to event was evaluated as the period from date of enrollment in the PURE study to the date of incident HF, death, or censoring, whichever came first. Thirty-day, 1-year, and 5-year case fatality was calculated using the time between the first HF event and the date of reported death. We compared age- and sex-adjusted HF prevalence and death rates using a general linear model. We illustrate all-cause mortality after incident HF in a Kaplan-Meier plot comparing subgroups of HICs and UMICs vs LMICs and LICs. To explore the relationship between baseline risk factors and time to incident HF, we used a multivariable cause-specific Cox shared frailty model to estimate the HR and 95% CI. We mutually adjusted for each of the 13 risk factors in addition to age, sex, and urban or rural area and included each community as a random intercept effect (ie, frailty term) in the model. Each of the risk factors was dichotomously categorized on the basis of a prespecified threshold level of risk.¹⁰ The fully

BOX 2 Understanding PAFs

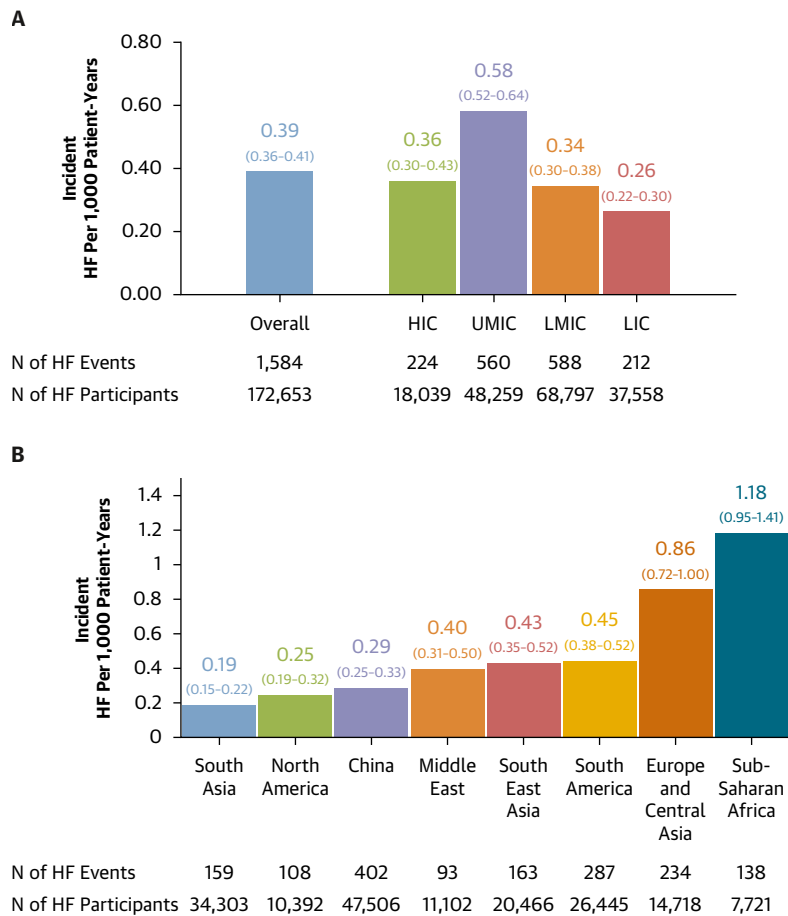
PAFs estimate the proportion of events in a population that could be theoretically avoided if a given risk factor were eliminated, assuming a causal relationship. PAFs depend on both the strength of association and the prevalence of the risk factor and will therefore vary across populations and settings. They reflect potential population-level impact rather than individual risk or the expected effect of a specific intervention.¹¹

adjusted model included 109,829 individuals (64%) without missing information on any of the included variables. The proportional hazards assumption was tested using Schoenfeld residuals and by visual inspection of log-log plots. We assessed the population attributable fraction (PAF) to understand the contribution of each risk factor for the development of HF in the overall population. We estimated the PAFs for individual risk factors using the average PAF approach of Eide and Gefeller¹¹ and the averisk package in R, developed by Ferguson et al.¹² Key concepts underlying PAFs are summarized in [Box 2](#). As in the Cox model, the risk factors were dichotomously categorized on the basis of a prespecified threshold level of risk.¹⁰ We derived average PAF estimates using a mutually adjusted model accounting for all candidate risk factors, age, sex, and urban or rural location; each risk factor was added sequentially to the model in all possible permutations, and then the average PAF from these permutations was calculated. In this approach, cumulative PAF estimates for groups of risk factors are additive and cannot exceed 100%. Analyses were done using SAS (SAS Institute) and R (R Foundation for Statistical Computing).

RESULTS

A total of 177,933 individuals from 712 communities consented to participate in phases 1 to 3 of the PURE study. We excluded 3,205 individuals who had not completed at least 1 round of follow-up ([Supplemental Figure 2](#)). At baseline, 2,075 of the participants (1.2%) had histories of HF. The age- and sex-standardized prevalence of HF was highest in UMICs (2.5%; 95% CI: 2.4%-2.6%), followed by LMICs (0.8%; 95% CI: 0.7%-0.8%), HICs (0.6%; 95% CI: 0.4%-0.7%), and LICs (0.5%; 95% CI: 0.4%-0.6%).

After excluding these 2,075 individuals, we estimated the incidence of HF among the remaining 172,653 individuals. Of these, 18,039 (10%) were from HICs, 48,259 (28%) from UMICs, 68,797 (40%) from LMICs, and 37,558 (22%) from LICs. During a median follow-up period of 14.9 years (Q1-Q3: 12.1-16.5 years), 1,587 individuals reported first unrefuted HF events (3% were refuted during review). Of these, 40% were classified as definite, 20% as probable, and 40% as possible HF. In HICs, the proportions of cases classified as definite, probable, and possible were 87%, 4%, and 9% respectively; in UMICs, these proportions were 33%, 20%, and 46%, respectively; in LMICs, they were 36%, 23%, and 42%, respectively; and in LICs, they were 18%, 28%, and 55%, respectively ([Supplemental Figure 1](#)).

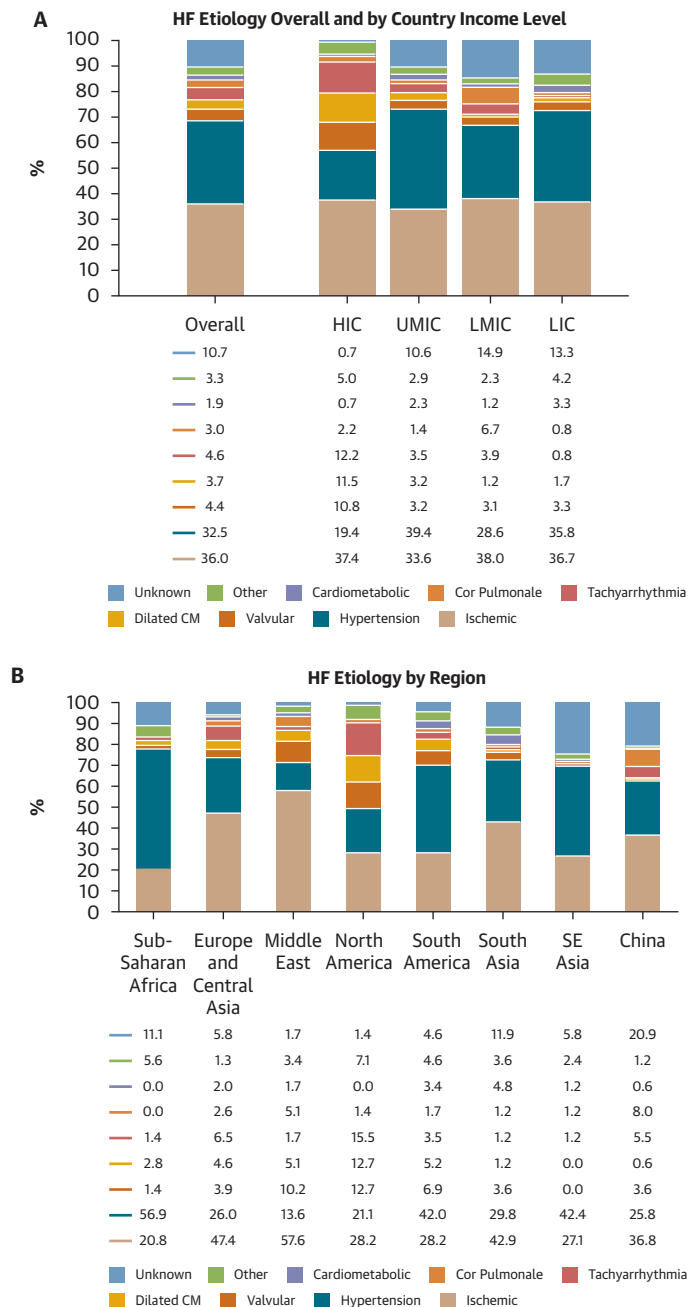
FIGURE 1 Age- and Sex-Standardized Incidence Rates of Unrefuted HF

(A) Overall and stratified by income setting. (B) Stratified by geographic region. HF = heart failure; HIC = high-income countries; LIC = low-income countries; LMIC = lower middle-income countries; UMIC = upper middle-income countries.

The overall age- and sex-standardized rate of incident HF was 0.39 (95% CI: 0.36-0.41) per 1,000 person-years (**Figure 1A**). It was highest in UMICs (0.58; 95% CI: 0.52-0.64), followed by HICs (0.36; 95% CI: 0.30-0.43), then LMICs (0.34; 95% CI: 0.30-0.38), and then in LICs (0.26; 95% CI: 0.22-0.30) (**Figure 1A**). Geographic regions with the highest incidence of HF were sub-Saharan Africa (1.18; 95% CI: 0.95-1.41) and Europe and Central Asia (incidence rate: 0.86; 95% CI: 0.72-1.00); regions with the lowest incidence were North America (0.25; 95% CI: 0.19-0.32) and South Asia (0.19; 95% CI: 0.15-0.22) (**Figure 1B**).

Ischemic HF and hypertensive HF each accounted for about one-third of incident HF cases, while the remaining comprised a combination of other less frequent etiologies (**Figure 2A**). For 11% of cases, it was not possible to assign etiology, because of a lack

of information. Ischemic heart disease and hypertension were the 2 most common etiologies among participants from all countries at different income levels (**Figure 2A**). Valvular heart disease, dilated cardiomyopathy, and tachyarrhythmia were more common etiologies in HICs compared with the other income groups. Ischemic heart disease and hypertension were also the most common etiologies in all 8 geographic regions, with ischemic etiology being more dominant in the Europe and Central Asia, the Middle East, and South Asia, while hypertensive etiology was more common in South America, Southeast Asia, and sub-Saharan Africa (**Figure 2B**). Left ventricular ejection fraction (LVEF) data were available in 187 of the incident HF cases (12%); of these, 91 (47%) were found to have LVEFs <40%, 31 (17%) had LVEFs of 41% to 50%, and 71 (35%) had LVEFs >50% ($P < 0.001$).

FIGURE 2 HF Etiology

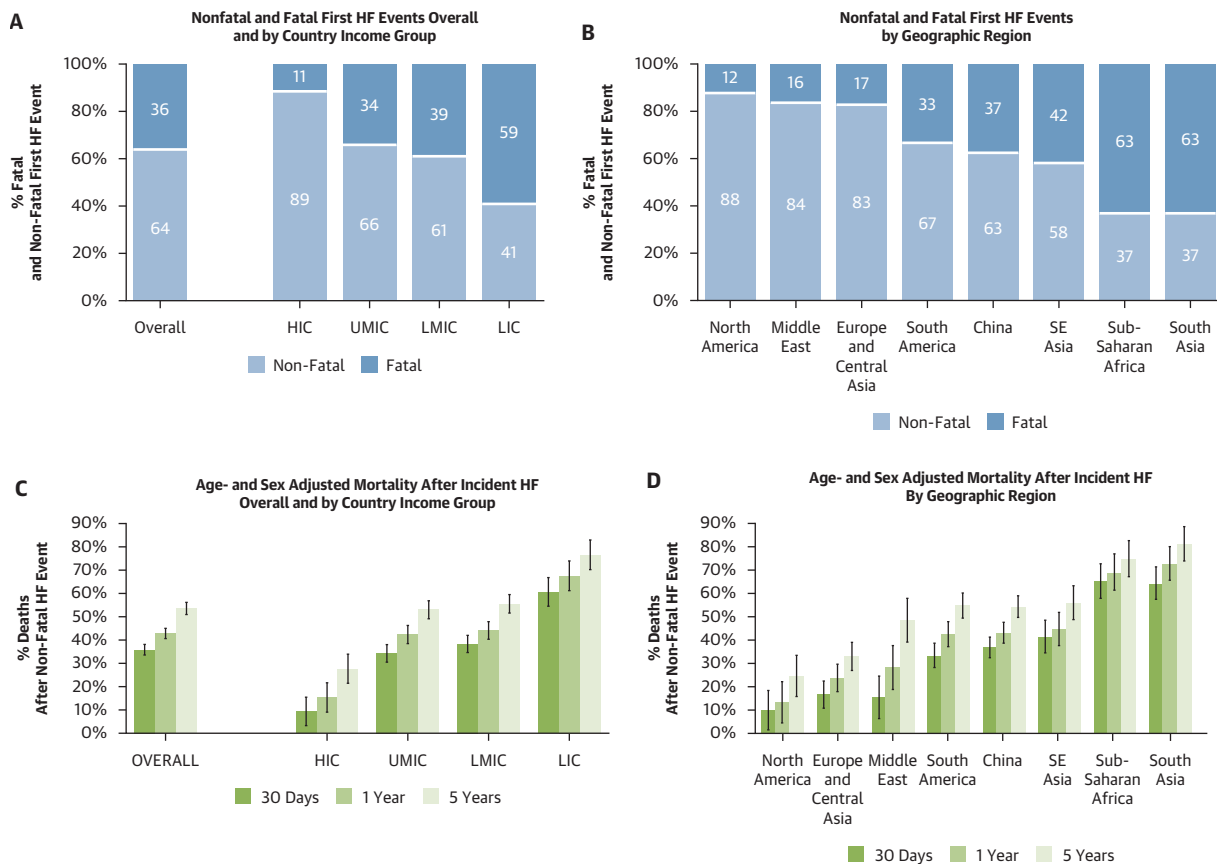
(A) Overall and by country income group. (B) By geographic region. "Other" etiologies are detailed in [Supplemental Appendix A4](#). CM = cardiomyopathy; other abbreviations as in [Figure 1](#).

Overall, 36% of the first HF events presented as fatal (ie, death within 30 days of first presentation). Proportions of fatal first HF events were lowest in HICs (11%) and highest in LICs (59%) ([Figure 3A](#)).

Fatal first HF events were lowest in North America (12%) and Europe and Central Asia (17%) and highest in South Asia (63%) and sub-Saharan Africa (63%) ([Figure 3B](#)). Age- and sex-standardized case fatality rates after a first HF presentation were 36% (95% CI: 34%-38%) at 30 days, 43% (95% CI: 40%-45%) at 1 year, and 54% (95% CI: 51%-56%) at 5 years ([Figure 3C](#)). The highest age- and sex-adjusted 30-day case fatality rate was seen in LICs (61%; 95% CI: 54%-67%). It was lower in LMICs (38%; 95% CI: 35%-42%), followed by UMICs (34%; 95% CI: 31%-38%), and lowest in HICs (9%; 95% CI: 3%-15%) ($P < 0.0001$). The highest 1-year case fatality rate was seen in LICs (68%; 95% CI: 61%-74%). It was lower in LMICs (44%; 95% CI: 40%-48%), followed by UMICs (43%; 95% CI: 39%-46%), and lowest in HICs (15%; 95% CI: 9%-22%) ($P < 0.0001$). Similar trends were seen for 5-year case fatality rate; it was highest in LICs (77%; 95% CI: 70%-83%), followed by LMICs (56%; 95% CI: 52%-59%) and UMICs (53%; 95% CI: 49%-57%), and lowest in HICs (28%; 95% CI: 21%-34%) ($P < 0.0001$). Among regions, the highest age- and sex-adjusted 30-day case fatality rates were in sub-Saharan Africa (65%; 95% CI: 58%-73%) and South Asia (64%; 95% CI: 57%-71%) and lowest in Europe and Central Asia (17%; 95% CI: 11%-22%) and North America (10%; 95% CI: 1%-18%) ($P < 0.0001$) ([Figure 3D](#)). Similarly, the highest 5-year rates were observed in South Asia (81%; 95% CI: 74%-89%) and sub-Saharan Africa (75%; 95% CI: 67%-83%) and lowest in Europe and Central Asia (33%; 95% CI: 27%-39%) and North America (25%; 95% CI: 16%-33%) ($P < 0.0001$) ([Figure 3D](#)). All 30-day, 1-year, and 5-year case fatality rates are tabulated in [Supplemental Table 2](#). The Kaplan-Meier curve ([Figure 4](#)) displays time to all-cause mortality after an incident HF event in individuals grouped by HICs and UMICs vs LMICs and LICs. Observed HF detection pathways and outcomes are conceptually summarized by country income level in the [Central Illustration](#).

RISK FACTORS AND PAFs FOR HF DEVELOPMENT.

[Table 1](#) compares baseline characteristics of participants who reported new-onset HF events during follow-up with those of participants who did not develop HF. Participants with incident HF were on average 8.3 years older than those without HF (58.6 ± 8.9 years vs 50.3 ± 9.9 years), were more commonly men (50.4% vs 41.1%), and had lower education levels (54.1% vs 41.6% with primary school or less). Participants who developed HF were more often overweight (28.1% vs 18.1% with body mass index ≥ 30 kg/m², 63.4% vs 51.3% with abdominal obesity [waist-to-hip ratio >0.9 in men and >0.85 in women]) and more often had hypertension (67.7% vs

FIGURE 3 Proportions of Nonfatal and Fatal First HF Events

(A) Overall and by country income group. (B) By geographic region. (C) Age- and sex-standardized 30-day, 1-year, and 5-year case fatality rates in incident HF cases overall and by country income group. (D) Age- and sex-standardized 30-day, 1-year, and 5-year case fatality rates in incident HF cases by geographic region. Comparisons across country income groups and geographic regions were highly statistically significant ($P < 0.01$). Abbreviations as in [Figure 1](#).

38.7%), ischemic heart disease (14.6% vs 3.4%), diabetes (25.1% vs 10.7%), and depression (13.9% vs 11.6%) at baseline. They also more often reported smoking (current or former; 42.9% vs 31.5%) and lower level of physical activity (24.7% vs 17.9% according to METs).

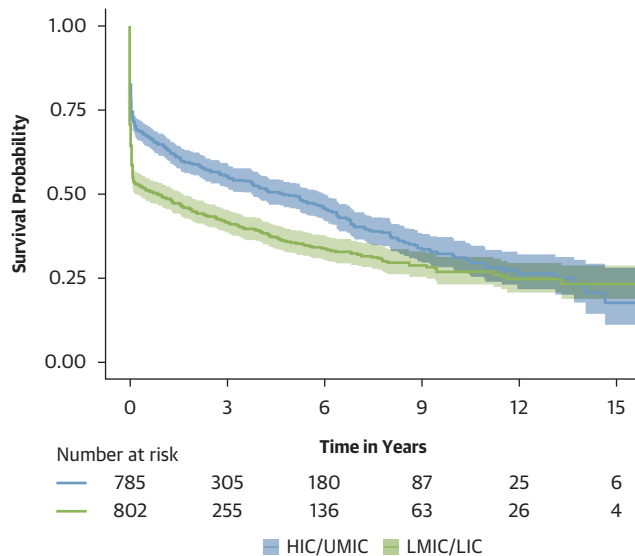
The strongest risk factors associated with incident HF ([Supplemental Table 3](#)) were ischemic heart disease (HR: 2.76; 95% CI: 2.33-3.28), hypertension (HR: 1.91; 95% CI: 1.65-2.20), diabetes (HR: 1.89; 95% CI: 1.63-2.20), and tobacco use (HR: 1.67; 95% CI: 1.41-1.99). Other independent risk factors associated with a higher risk for developing HF were low education, presence of depressive symptoms, low level of physical activity, poor diet quality, and physical weakness (low grip strength). Excessive alcohol use, indoor air pollution, and increased non-high-density lipoprotein cholesterol were not independently

associated with a higher risk for HF. We further explored the risk factors for HF stratified into HICs and UMICs vs LMICs and LICs, finding largely similar trends ([Supplemental Table 3](#)).

The 13 studied modifiable risk factors contributed to approximately 71.7% of the PAF for incident HF ([Figure 5](#)). The largest PAFs were attributable to hypertension (24.6%), low education (7.6%), tobacco use (6.5%), ischemic heart disease (6.1%), and diabetes (5.9%). Indoor air pollution and poor diet contributed to 4.4% and 4.3%, respectively, while other risk factors each contributed to <3.3% of the PAF for HF.

DISCUSSION

To our knowledge, this is the first multinational cohort study to report on HF incidence in

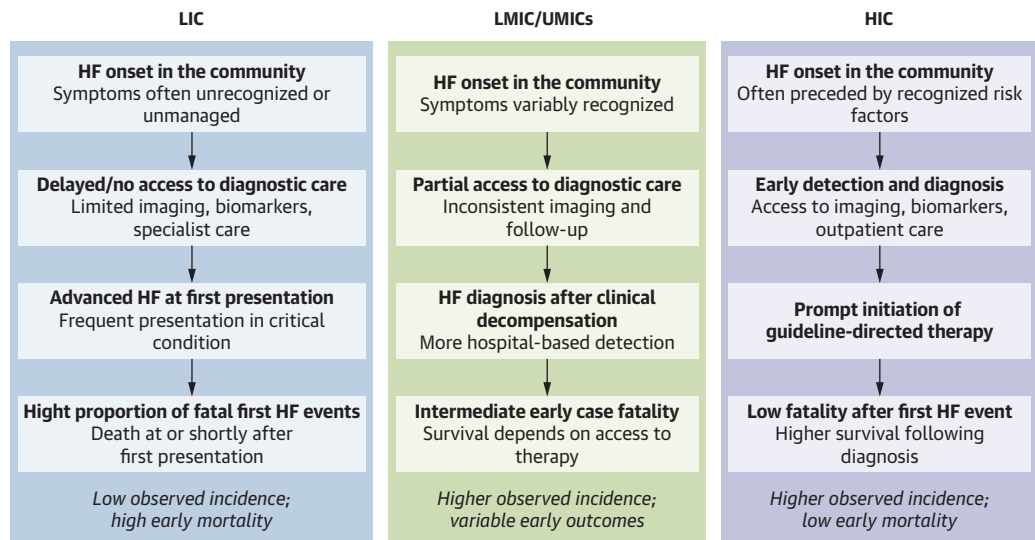
FIGURE 4 Kaplan-Meier Survival Plot Displaying Time From First HF Event to All-Cause Death in Individuals From HICs and UMICs vs LMICs and LICs

Abbreviations as in Figure 1.

communities from HICs, UMICs, LMICs, and LICs, and in multiple world regions, using standardized data collection. We identified 3 key findings. First, the highest incidence of HF was observed in UMICs and in the regions of sub-Saharan Africa and Europe and Central Asia. Second, mortality after incident HF is substantially higher in LICs and MICs compared with HICs. Third, more than two-thirds of HF cases are attributable to potentially modifiable risk factors, with hypertension being the single largest population-level risk factor for the development of HF globally.

We observed a 2-fold difference in standardized HF incidence among countries grouped by income level, with the highest incidence in UMICs (0.58), followed by HICs (0.36), then LMICs (0.34), and then LICs (0.26). Prior attempts to compare HF incidence in different parts of the world have largely relied on comparing summary information from different studies, with vastly different population characteristics and methods of ascertaining HF. These prior studies have suggested that HF incidence is higher in HICs compared with UMICs and that there are few data from LMICs or LICs to make reliable

CENTRAL ILLUSTRATION Conceptual Illustration of HF Detection Pathways and Early Outcomes Across Income Settings



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Differences in health literacy, access to care, diagnostic capacity, and treatment influence the timing of heart failure (HF) detection and survival after first presentation. Lower income settings experience underdiagnosis to a higher degree and a higher proportion of fatal first HF events. HIC = high-income countries; LIC = low-income countries; LMIC = lower middle-income countries; UMIC = upper middle-income country.

TABLE 1 Baseline Characteristics of the Population With Incident HF vs No HF

	No HF (n = 171,066)	Incident HF (n = 1587)
Age, y	50.3 ± 9.9	58.6 ± 8.9
Men	70,342 (41.1)	799 (50.4)
Education		
Primary or less	70,895 (41.6)	854 (54.1)
Secondary	64,338 (37.7)	451 (28.6)
Trade/college/university	35,347 (20.7)	273 (17.3)
Rural residence	80 655 (47.2)	813 (51.2)
Tobacco use		
Current	34,162 (20.2)	381 (24.2)
Former	19,255 (11.4)	293 (18.6)
Alcohol use		
Never	117,665 (70.5)	1,004 (65.1)
PURE diet score ^a		
>4	20,396 (13.2)	180 (12.2)
3 or 4	60,148 (38.9)	547 (37.1)
≤2	73,990 (47.8)	749 (50.7)
Physical activity: MET-min/wk		
Low (<600)	28,226 (17.9)	355 (24.7)
Moderate (600-3,000)	58,797 (37.2)	508 (35.3)
High (≥3,000)	70,976 (44.9)	576 (40.0)
Hypertension	66,126 (38.7)	1,070 (67.7)
Ischemic heart disease	5,913 (3.4)	232 (14.6)
Diabetes	18,251 (10.7)	399 (25.1)
Depression	19,600 (11.6)	218 (13.9)
BMI group		
<25 kg/m ²	76,624 (47.7)	580 (39.0)
25-30 kg/m ²	55,060 (34.3)	488 (32.8)
≥30 kg/m ²	29,072 (18.1)	418 (28.1)
WHR >0.9 (men) or >0.85 (women)	82,394 (51.3)	948 (63.4)
Non-HDL, mmol/L	3.70 ± 1.0	3.80 ± 1.2
Grip strength, kg	30.3 ± 11.2	29.4 ± 11.2
Indoor air pollution	43,906 (26.2) ^b	440 (28.1) ^b

Values are mean ± SD or n (%). P values are <0.02 unless otherwise stated.

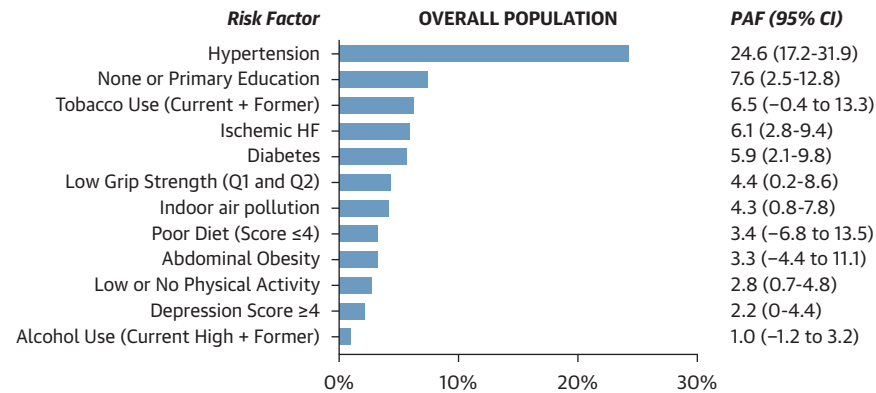
^aRange: 0 to 6; lower scores represent a lower quality diet. ^bNot significant.

BMI = body mass index; HDL = high-density lipoprotein; HF = heart failure; PURE = Prospective Urban Rural Epidemiology; WHR = waist-to-hip ratio.

comparisons.¹³⁻²⁴ Using a standardized population, our observation that HF incidence is currently highest in UMICs is a novel finding, indicating that in this group of countries, strategies to reduce the growing burden of HF are of particular importance. We also observed that the standardized rates of HF observed across the PURE study were lower than what has been observed in prior studies, which report HF incidence rates of 2 to 7 cases per 1,000 person-years in Western HICs.¹³⁻²¹ Higher incidence rates of HF reported in these studies compared with PURE are likely related to several factors, including the inclusion of healthier older populations, and many of these studies were conducted using administrative

databases with HF ascertained via diagnostic codes. Although this approach is useful in HICs, it does not allow robust comparisons with MICs or LICs, where such administrative databases largely do not exist. In PURE, HF cases were ascertained if reported by participants, through hospitalizations or deaths, so that our cohort likely recorded more symptomatic and severe HF cases. This approach would reduce the absolute number of identified HF cases but provides more comparable data across groups of countries at different income levels than prior studies.

In the PURE study, mortality after incident HF varied substantially by country income level and by geographic region. The largest relative difference in mortality occurred within the first 30 days; a striking 59% died in LICs, 39% in LMICs, 34% in UMICs, and only 11% in HICs, indicating an up to 5-fold difference in first HF events presenting as fatal between countries at different income levels. These relative differences in case fatality decreased over time but remained highest in LICs and lowest in HICs at 5 years. Extremely high mortality rates after incident HF events were seen in South Asia and sub-Saharan Africa, regions where most countries included in PURE are either LICs or LMICs. Similar trends have been observed in multinational HF registries, which have reported higher mortality in lower income settings.^{4,25,26} However, absolute 1-year case fatality rates from LMICs and LICs in PURE are substantially greater than the annual rates observed in these registries.^{4,25,26} Data from PURE thus indicate that community-based HF mortality in LICs and LMICs is considerably higher than previously estimated from registry data, which generally include more selective HF populations.²⁷ Moreover, in LICs and to some extent also in LMICs and UMICs, the high mortality is driven largely by the individuals' presenting with fatal HF events. This may be due to several factors, including individuals in these countries presenting later with HF or poor management of common HF risk factors. In other words, it may be that the observed pattern of low HF incidence rates and high early case fatality in LICs and LMICs does not reflect a genuinely lower burden of HF but rather a consequence of limitations in health system access and HF detection. This has an important implication: reducing mortality in HF in these countries will require not only improvements in chronic HF management but also a greater focus on improved treatment of upstream risk factors and care of acute conditions that lead to HF, such as the management of myocardial infarction and arrhythmias, outside of HICs. Moreover, these findings highlight the need for the implementation of contextualized strategies to

FIGURE 5 PAFs With Their 95% CIs for 13 Modifiable Risk Factors for Incident HF in the Population at Risk

Estimates for individual risk factors were truncated at a lower limit of 0, as this is the lowest threshold to demarcate a relationship with increased risk. Cholesterol level did not contribute to the risk for incident HF and is not shown. HF = heart failure; PAF = population attributable fraction; Q = quartile.

improve clinical outcomes following HF in low-resource settings.^{4,25-28}

Finally, our data suggest that the development of HF in the global setting is related largely to modifiable risk factors. Ischemic and hypertensive heart disease were the leading causes of HF, accounting for nearly two-thirds of cases. Similarly, we observed that 71% of HF cases in our overall cohort were related to potentially modifiable risk factors, and 51% were related to 5 factors: hypertension, low education, tobacco use, ischemic heart disease, and diabetes. Hypertension was by far the largest population-level risk factor for HF development globally, itself accounting for one-quarter of the PAF for HF. These findings are consistent with the impact of hypertension as the largest risk factor for CVD globally and emphasize the need for continued strategies to improve global efforts to prevent and treat hypertension. Low education was the second leading global risk factor, contributing to 7.6% of the PAF of HF after accounting for other cardiometabolic and behavioral risk factors. It is likely that low education contributes to HF development through broader socioeconomic effects and access to health care, and given its relatively high contribution to HF development globally, there is a need to better understand how HF risk can be mitigated in groups with low socioeconomic status. At the population level, the predominant risk factors for HF continue to largely align with those of CVD in general, suggesting that CVD prevention strategies aimed at detecting and

controlling hypertension, targeting obesity and cardiometabolic risk factors and tobacco use, and improving outcomes in low education groups will have benefits for the prevention of HF.

STUDY LIMITATIONS. The present analysis of the PURE study included 25 countries from a broad range of geographic and health resource settings but lacked representation from West Africa, North Africa, and Oceania. Recruitment occurred in specific communities within participating countries and therefore may not necessarily be nationally representative. However, PURE remains one of the most broadly applicable global population-based studies, as we have included a large population from >1,000 urban and rural communities from the 25 included countries. Although adjudication minimized the misdiagnosis of HF, some events may have been missed because of our ascertainment methods, which would generally underestimate the number of cases identified compared with administrative data sets using International Classification of Diseases codes. Furthermore, comparability across income settings is affected by large differences in the proportion of HF cases classified as definite, probable, or possible. However, at the moment, our approach is the only practical strategy to compare data on HF across countries at different income levels, as robust administrative databases that identify HF are still available in only a few Western HICs. Finally, further characterization by HF phenotype was not

possible, because of low availability of echocardiographic data. This may also have influenced the assigned etiology.

CONCLUSIONS

HF incidence varies widely among country-income level groups and geographic regions. Strategies are needed to mitigate the large disparities in HF-related mortality occurring globally. Effective prevention and management strategies should prioritize modifiable risk factors, especially hypertension.

DATA AVAILABILITY Unpublished data related to this study are not available at present for public use.

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KEY WORDS global, heart failure, incidence, mortality, population, prevalence

APPENDIX For supplemental methods, a complete list of the PURE study investigators, and supplemental tables and figures, please see the online version of this paper.